

Rakesh Rosan Pradhan

DevOps Engineer,

Gebbs Technologies Limited

Email: btechece014@gmail.com

Mobile +91-9437702221

Career Objective:

Results-driven IT professional with over 6+ years of extensive IT experience and 4+ years in DevOps, Infrastructure Automation, and Configuration Management across DevOps, Linux, and AWS/GCP environments. I seek to leverage my expertise in designing, deploying, and managing scalable, secure, and efficient cloud-based solutions to contribute to the success of a forward-thinking organization. Passionate about continuous integration, automation, and optimizing IT workflows to deliver high-quality, cost-effective infrastructure solutions that drive business growth and innovation.

Professional Summary:

- ✓ Accomplished Cloud & DevOps Engineer with over 4 years of experience in architecting, automating, and optimizing scalable cloud environments using AWS/GCP and industry-standard DevOps practices. Proficient in designing high-availability, fault-tolerant architectures with AWS services like IAM, VPC, EC2, S3, ELB, Auto Scaling, and Lambda. Skilled in CI/CD pipeline management with Jenkins, GitHub, and GitLab, along with expertise in artifact management using JFrog. Strong hands-on experience in infrastructure automation using Terraform, containerization with Docker, and orchestration with Kubernetes, including Helm for efficient application deployments.
- ✓ Expert in cloud infrastructure monitoring with Prometheus, Grafana, and AWS CloudWatch, ensuring optimal system health and performance. Adept at cost optimization through resource rightsizing and automation. Experienced in managing RDS, VPC Peering, and web servers like Apache and Tomcat. Proficient in Windows and Linux environments, leveraging scripting and automation to enhance efficiency. Committed to implementing cutting-edge DevOps practices for streamlined deployment, reduced operational complexity, and improved scalability.

Work Experience:

- ✓ Working as Aws & DevOps Engineer in **GeBBs Technologies Limited**, Andheri-Kurla Road, Mumbai from March 2019 to present.

Technical/ Functional Summary:

CATEGORIES	TECHNOLOGY
Operating systems	RHEL6.7, Ubuntu 22.04, Windows
Version Control Tools	Git, GitHub, GitLab
CI/CD tools	Jenkins, GitHub Actions
Provisioning tool	Terraform, AWS CloudFormation
Configuration Management Tools	Ansible
Containerization Tool	Docker, Kubernetes
Scripting languages	Bash, Shell Scripting, Python.
Application server	Apache, Nginx
Container Orchestration	Kubernetes (K8s), EKS (Elastic Kubernetes Service)
Infrastructure as Code (IaC)	Terraform, Helm Charts, AWS CDK
Cloud Platform	AWS and GCP
Databases	MySQL and Mongo DB
Monitoring & Logging	Prometheus, Grafana, AWS CloudWatch
Agile & Project Management	Jira, ServiceNow

Projects Summary:

Project #1

Project Name : Three Tire App on AWS EKS Argo CD
Client : MUBI
Role : DevOps Engineer
URL : <https://mubi.com/>

Description:

Mubi is a British global over-the-top (OTT) service streaming platform, production company and film distributor. MUBI produces and theatrically distributes films by emerging and established filmmakers, which are exclusively available on its platform. The catalogue includes world cinema films, such as art houses, documentaries and independent films. Additionally, it publishes Notebook, a film criticism and news publication, and provides weekly cinema tickets to selected new-release films through MUBI GO.

Objective: The objective of this project is to design, develop, and deploy a **scalable OTT streaming platform** using a **three-tier architecture** on **AWS EKS with Argo CD for GitOps-based automation**. The **frontend (React.js)** delivers a seamless user experience, the **backend (Node.js + Express.js)** manages authentication (JWT/Auth0/Firebase), streaming (AWS MediaLive/FFmpeg), and subscriptions (Stripe/Razorpay), while the **database layer (MongoDB Atlas)** handles user data and metadata. AWS S3 and CloudFront ensure optimized content delivery, while Redis caching improves performance. CI/CD is automated with Argo CD, and monitoring is powered by Prometheus, Grafana, and the ELK stack. The goal is to create a **resilient, scalable, and high-performance OTT platform** with low-latency video streaming and efficient resource utilization.

Project Overview:

1. Frontend (React.js - Presentation Layer) on AWS EKS

- **Justification:** Hosting the React.js frontend in a Kubernetes pod allows for auto-scaling and load balancing.
- **Deployment Approach:**
 - **Containerization:** React app is containerized using Docker.
 - **Ingress Controller:** AWS ALB Ingress for routing requests.
 - **CDN Integration:** CloudFront caches static content to reduce latency.
 - **Argo CD:** Automatically deploys frontend updates from GitHub/GitLab.

2. Backend (Node.js - Application Layer) on AWS EKS

- **Justification:** A Node.js backend running on Kubernetes provides elasticity and microservices orchestration.
- **Deployment Approach:**
 - **Containerization:** Node.js backend runs in scalable EKS pods.
 - **Service Discovery:** Exposed via Kubernetes Services (ClusterIP for internal, LoadBalancer for external APIs).
 - **Authentication & Security:**
 - JWT-based authentication with Passport.js or Firebase Auth.
 - API Gateway (AWS API Gateway or Kong) for secure access.
 - **Streaming Services:** AWS MediaLive or FFmpeg integrated via backend APIs.
 - **Caching Layer:** Redis deployed as a stateful set in EKS to reduce database load.
 - **Argo CD:** Manages application deployments and rollbacks efficiently.

3. Database (MongoDB - Data Layer) on AWS

- **Justification:** A managed NoSQL database like **MongoDB Atlas** ensures high availability and automatic scaling.
- **Deployment Approach:**
 - **MongoDB Atlas:** Managed NoSQL database for metadata, user profiles, and subscriptions.
 - **AWS S3:** For storing video content and thumbnails.
 - **CDN (CloudFront):** Speeds up video content delivery globally.
 - **Backup & Recovery:** AWS Backup configured for disaster recovery.

Additional Technologies & Justification

1. CI/CD with Argo CD

- Argo CD enables GitOps-style deployments, ensuring that EKS clusters remain synchronized with Git repositories.
- Supports automated rollbacks and progressive deployments (Blue-Green, Canary).

2. Containerization with Docker & Kubernetes

- EKS ensures auto-scaling and self-healing for all microservices.
- Helm charts manage deployments and configurations.

3. Monitoring & Logging

- **Prometheus & Grafana:** Monitor application performance and resource utilization.
- **ELK Stack:** Centralized logging for backend services.
- **AWS CloudWatch:** Integrated with EKS for real-time monitoring.

Project #2

Project Name : Automated Deployment & Monitoring for a CCTV Camera Surveillance System
Client : Hikvision, China
Role : DevOps Engineer
URL : <https://www.hikvision.com/en/>

Environment :

- GCP (EC2, S3, RDS, Kinesis, CloudWatch, Lambda, VPC, ELB, IAM)
- Jenkins, GitHub, Terraform, Ansible
- Docker, Kubernetes (EKS), Helm
- Prometheus, Grafana
- Bash, Python,

Description:

Developed and implemented a scalable **CCTV Camera Monitoring System** infrastructure using **AWS, Docker, Kubernetes**, and **Jenkins**. The system handles deploying and managing hundreds of CCTV cameras, ensuring reliable and real-time video streaming, data storage, and monitoring.

Roles & Responsibilities:

- ✓ Designed and automated cloud infrastructure provisioning using Terraform on GCP, enabling scalable and version-controlled deployments of resources like EC2, VPC, S3, RDS, and ELB. Leveraged Amazon S3 for efficient video storage from CCTV cameras, implementing lifecycle policies for cost-effective archival in Amazon Glacier. Built a CI/CD pipeline with Jenkins and GitHub for seamless application updates, ensuring rapid deployment of new camera software versions. Containerized monitoring applications with Docker and deployed them on Kubernetes (EKS) for scalable management, rolling updates, and health checks across multiple locations.
- ✓ Implemented real-time video streaming and processing with GCP Kinesis Video Streams and Lambda for motion detection alerts. Set up Prometheus, Grafana, CloudWatch, and Datadog for proactive monitoring and log aggregation. Enhanced security with IAM policies, AWS Secrets Manager, and SSL encryption. Automated remote software updates using Ansible and Jenkins, optimized costs with Auto Scaling groups and Spot Instances, and ensured high availability through multi-AZ deployments and cross-region replication for disaster recovery.

Project #3

Project Name : End-to-End Deployment on AWS EKS Cluster
Client : Insight TV (OTT), Amsterdam, North Holland
Role : DevOps Engineer
URL : <https://watch.insight.tv/>

Description:

INSIGHT TV is the leading Millennial and Gen-Z global broadcaster, distributor, and channel operator, and is available on over 178 platforms, across 56 Countries, in 12 languages. Based in Amsterdam, INSIGHT TV focuses on real-life, story-driven content across multiple genres and verticals. INSIGHT TV's linear flagship channels include INULTRA (4K UHD) and INPLUS (HD). as well as a suite of digital channels that include, INFAST (Lifestyle), IN TROUBLE (Action Sports), IN WONDER (Science & Technology), IN WILD (Nature & Wildlife co-venture with Off the Fence), and IN TRAVEL.

Objective: Led the deployment of a three-tier application on AWS EKS, utilizing GitOps with ArgoCD to automate and streamline the delivery process.

Project Overview:

This project provides an automated deployment pipeline for a three-tier application, utilizing a GitOps approach to manage deployments.

- **Infrastructure Provisioning:** Utilized Terraform to create and manage the AWS infrastructure, including the EKS cluster, VPCs, subnets, and security groups.
- **CI/CD Pipeline:** Configured Jenkins to build Docker images from the source code and push them to a Docker registry. Jenkins also triggers ArgoCD to sync and deploy or update the application on the EKS cluster.
- **Monitoring and Alerting:** Leveraged Prometheus to collect metrics from the EKS cluster and used Grafana for visualization. Configured alerts to notify the team via email in case of critical issues.

Key Achievements:

- **Infrastructure Automation:** Reduced setup time by 30% using Terraform for AWS provisioning.
- **CI/CD Pipeline:** Achieved a 95% deployment success rate with zero downtime by implementing a Jenkins-driven CI/CD pipeline.
- **Monitoring Setup:** Improved issue detection time by 40% with Prometheus and Grafana, enabling faster response to critical alerts.

Impact: Delivered a 99.9% deployment success rate, improved scalability, and reduced manual efforts by 90%, significantly enhancing operational efficiency and system reliability.

Tools Used:

- **Linux:** Development and deployment scripting.

- **GitHub:** Version control and GitOps repository hosting.
- **Docker:** Containerization of application services.
- **Jenkins:** Continuous Integration and Continuous Deployment (CI/CD) pipeline.
- **Kubernetes (AWS EKS):** Orchestration and management of containerized applications.
- **Terraform:** Infrastructure as Code (IaC) to provision AWS resources and Kubernetes clusters.
- **Prometheus:** Monitoring system to collect metrics from Kubernetes clusters.
- **Grafana:** Visualization of monitoring data from Prometheus.
- **Mailing:** Notification setup for deployment status and monitoring alert

Previous Work Experience

Working as B Class Govt Engineer Contractor.

From: 2015 to 2018

Role: Contractor (Working in Govt of Odisha various Dept like Irrigation, Drainage, R&B, RD etc through e-Tender.

Personal Summary:

Academic Credentials:

- ✓ Bachelor of Technology in Electronics and Communication Engineering (ECE) in BPUT 2014 Bhubaneswar.
- ✓ +2 Science at SVM Autonomous College, Jagatsinghpur under CHSE 2010
- ✓ 10th at Purohitpur High School, Jagatsinghpur under BSE, Odisha 2008

PERSONAL DETAILS:

Date of Birth : 22/05/1993
 Father : Sukant Kumar Pradhan
 Languages Known : Odia, Hindi, English,
 Permanent Address : Tardapada, Jagatsinghpur, 754103

Declaration:

I hereby declare that the details and information given above are complete and true to the best of my knowledge

Date:

Yours Sincerely

Place:

Rakesh Rosan Pradhan